Proposed Research Title

A Multi-Resolution, Energy-Efficient Edge Audio Intelligence System for Continuous Low-Power Streaming Speech Recognition in Real-World Environments

Abstract

The growing demand for real-time speech understanding in mobile and embedded devices requires audio intelligence systems capable of operating continuously under strict power and computational constraints. Applications such as smart assistants, hearing augmentation, meeting transcription, and environmental monitoring require accurate speech recognition despite noise, reverberation, multiple speakers, and diverse acoustic environments. However, conventional cloud-based automatic speech recognition (ASR) pipelines are unsuitable for always-on operation due to latency, privacy concerns, and energy consumption. This project proposes the design and implementation of a **mobile-grade edge audio intelligence system** capable of continuous low-power listening while maintaining high transcription accuracy across diverse real-world conditions.

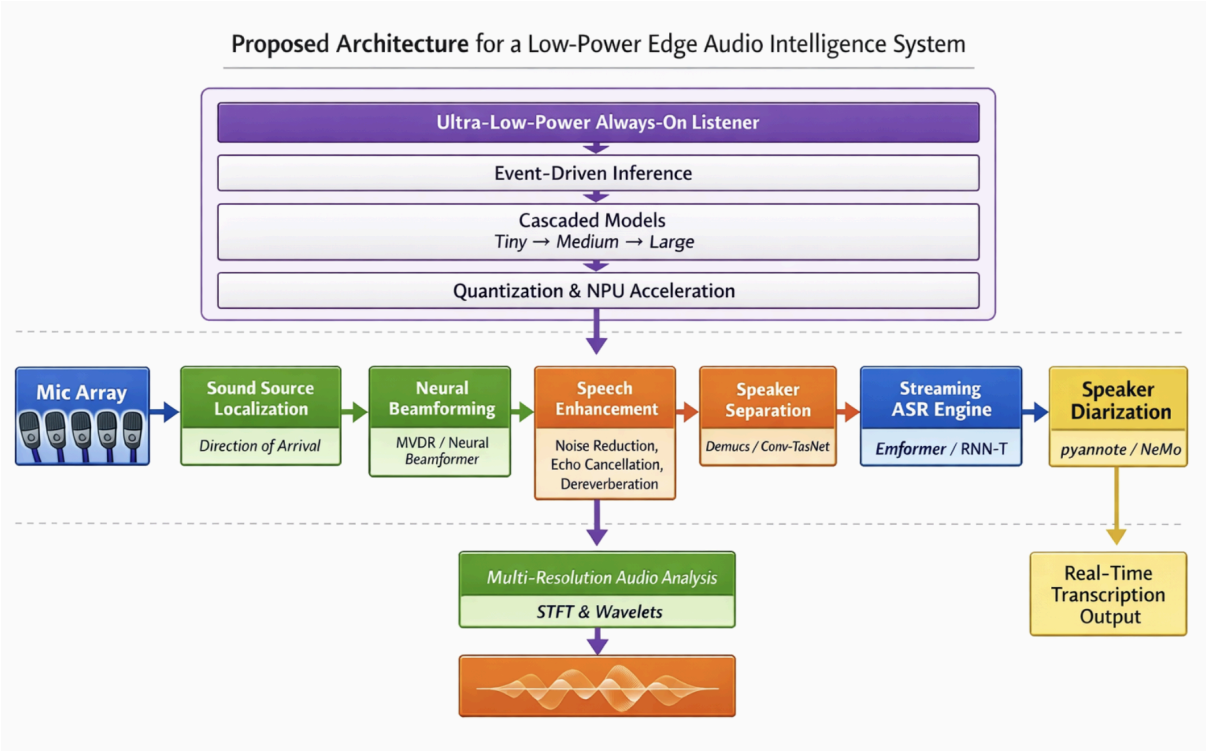
The proposed system integrates several advanced signal processing and machine learning modules into a unified architecture optimized for edge deployment. The front-end stage utilizes **multi-microphone beamforming and sound source localization** to capture spatially focused speech signals and suppress environmental interference. The captured audio is then processed through **speech enhancement and noise suppression modules**, including neural speech separation and dereverberation models, to improve speech clarity under noisy and reverberant conditions. A **multi-resolution audio processing framework** is explored to analyze acoustic signals across different temporal and spectral scales, improving robustness and interpretability in complex environments.

To enable continuous operation with minimal energy consumption, the system incorporates an **ultra-low-power always-on listener** responsible for speech activity detection, acoustic scene classification, and preliminary sound analysis. This stage triggers higher-capacity models only when relevant speech events are detected, enabling **event-driven inference and cascaded model execution**. For real-time transcription, the system employs a **streaming transformer-based ASR architecture optimized for low-latency inference**, combined with model compression techniques such as quantization and knowledge distillation for efficient edge deployment.

Additionally, the system integrates **speaker diarization and acoustic context awareness**, allowing it to identify multiple speakers and dynamically adapt its signal processing strategies based on environmental conditions such as classrooms, meetings, or noisy public spaces. The overall architecture is designed to operate continuously on edge devices while minimizing power consumption and computational load.

The primary objective of this research is to develop a **scalable, interpretable, and energy-efficient speech intelligence framework capable of accurate real-time speech transcription and audio scene understanding on mobile or embedded hardware**. By combining multiresolution signal processing with modern deep learning architectures and

efficient system design, this work aims to push the boundaries of **always-on edge speech recognition systems**, enabling reliable audio intelligence in practical real-world deployments.



The Core Technical Challenge

Your project addresses the challenge of building a system that simultaneously achieves:

1. Robust perception

Accurately interpret speech despite:

- noise
- Reverberation
- overlapping speakers
- acoustic interference

2. Low power consumption

The system must run continuously without draining battery.

3. Real-time operation

Speech must be transcribed with **minimal latency**.

4. Edge deployment

Processing must occur **on-device**, avoiding cloud dependency

Key Constraints of Edge Audio Systems

Understanding constraints is essential before designing algorithms.

3.1 Hardware Constraints

Mobile devices typically have:

Resource	Limitation
Microphones	2–4 channels
Memory	limited RAM
Compute	mobile CPU / DSP / NPU
Battery	limited energy budget

These constraints strongly influence algorithm selection.

3.2 Acoustic Constraints

Real-world audio environments introduce:

Noise

Sources include:

- Traffic
- crowd chatter

- wind
- mechanical sounds

Reverberation

Reflections from walls cause temporal smearing of speech signals.

Overlapping speakers

Multiple speakers may talk simultaneously.

Directional ambiguity

The system must determine **where sound originates**.

Prior research consistently shows that speech processing systems must operate effectively across **wide signal-to-noise ratio (SNR) ranges**, complex reverberant environments, overlapping speech conditions, and strict energy and latency constraints.

Benchmark studies such as the CHiME challenge demonstrate that real-world speech recognition systems must function across SNR levels ranging approximately from **−6 dB to +9 dB**, while enhancement benchmarks such as the DNS Challenge extend this range to roughly **−5 dB to +20 dB**, reflecting realistic noise conditions found in environments such as streets, cafes, and public transport. These findings indicate that robust speech recognition requires effective front-end enhancement and spatial filtering to maintain performance below +5 dB SNR, where conventional systems typically degrade rapidly.

Reverberation further complicates real-world deployment. Experimental evaluations in room acoustics and CHiME simulations show that typical indoor environments exhibit reverberation times (RT60) between approximately **0.2 s and 1.0 s**, with studies reporting significant increases in word error rate (WER) under higher reverberation conditions. Similarly, conversational corpora such as the AMI Meeting Corpus indicate that overlapping speech occurs in approximately **10–20% of real meeting scenarios**, necessitating reliable source separation and diarization modules.

Spatial processing constraints are also well documented. Recent low-power sound source localization studies report that practical beamforming systems typically require direction-of-arrival estimation errors below roughly **10°**, with deep learning-based localization methods achieving average angular errors of approximately **2–6°** under realistic conditions.

From a systems perspective, continuous on-device operation introduces strict energy and latency constraints. Studies on always-on speech processing report that ultra-low-power listening modules must operate within approximately **1–5 mW**, while full streaming ASR pipelines typically consume **200–500 mW**, making cascaded and event-driven architectures essential. Furthermore, human-interaction studies and streaming ASR benchmarks indicate that end-to-end latency must remain below approximately **300 ms** to maintain usability for real-time transcription and conversational interaction.

Together, these findings establish the practical design envelope for edge speech intelligence systems: operation across SNR ranges of roughly −5 to +20 dB, reverberation up to ~1 s RT60, frequent overlapping speech, strict localization accuracy requirements, and stringent energy and latency budgets. These constraints strongly motivate the use of spatial filtering, multiresolution signal processing, cascaded inference architectures, and efficient streaming ASR models in real-world deployments.

Best Sources (Use These)

1. Barker et al., “The CHiME Speech Separation and Recognition Challenge,” *Computer Speech & Language*, 2018.

2. Reddy et al., “The INTERSPEECH 2020 Deep Noise Suppression Challenge,” *INTERSPEECH*, 2020.

3. Carletta et al., “The AMI Meeting Corpus,” *International Journal of Speech Technology*, 2007.

4. Shi et al., “Emformer: Efficient Memory Transformer Based Acoustic Model for Low Latency Streaming Speech Recognition,” *INTERSPEECH*, 2021.

5. Grinstein et al., “Real-Time Sound Source Localization for Low-Power IoT Devices Based on Multi-Stream CNN,” *IEEE Sensors Journal*, 2022.

Constraint	Typical Range
SNR	−5 dB to +20 dB
RT60	0.2–1.0 s
Speaker overlap	10–30%
DoA error tolerance	<10°
Always-on power	<5 mW

Full ASR power	<500 mW target
Latency target	<300 ms
Edge model size	<50M params

System-Level Decomposition

To solve the problem systematically, the system is decomposed into functional modules.

This modular decomposition follows common practice in **speech processing research**.

System pipeline:

Audio Capture

↓

Sound Source Localization

↓

Beamforming

↓

Speech Enhancement

↓

Speaker Separation

↓

Streaming Speech Recognition

↓

Speaker Diarization

↓

Text Output

Each module solves a **specific signal processing challenge**.

5. Audio Capture and Spatial Understanding

Objective

Capture the **cleanest possible signal** from microphones.

Why this matters

Speech recognition accuracy is highly dependent on **input signal quality**.

Improving audio quality before ASR significantly improves final transcription accuracy.

5.1 Microphone Array Processing

Multiple microphones allow spatial filtering.

Principle:

Sound reaches microphones with **different time delays**.

By analyzing these delays, the system can estimate sound direction.

5.2 Sound Source Localization

Goal:

Estimate **direction of arrival (DoA)**.

Important because:

- allows spatial filtering
- enables speaker tracking

Classical Methods

GCC-PHAT

Generalized Cross-Correlation with Phase Transform.

Advantages:

- simple
- computationally efficient
- interpretable

Limitations:

- sensitive to noise
- reduced accuracy in reverberant environments.

Deep Learning Methods

Recent research uses neural networks for localization.

Example:

Multi-stream CNN localization.

Advantages:

- robust in noisy environments
- handles multiple sources.

Reference:

Grinstein et al., *Real-Time Sound Source Localization for Low-Power IoT Devices*.

6. Beamforming

Beamforming focuses the microphone array toward the speaker.

Concept

Combine signals from multiple microphones to enhance sound from a particular direction.

6.1 Delay-and-Sum Beamforming

Signals are delayed and summed so the target signal aligns.

Advantages:

- simple
- efficient.

Limitations:

- limited interference suppression.

6.2 MVDR Beamforming

Minimum Variance Distortionless Response.

Objective:

Minimize noise while preserving desired signal.

Advantages:

- better noise suppression
- widely used in speech systems.

6.3 Neural Beamforming

Neural networks estimate optimal spatial filters.

Advantages:

- adaptive
- robust.

Libraries:

ESPNet.

7. Speech Enhancement

After spatial filtering, the signal still contains noise and distortions.

Enhancement improves signal clarity.

Types of distortions

Noise

Background sound contamination.

Reverberation

Reflections causing temporal smearing.

Echo

Signal reflections from speakers.

Enhancement Approaches

Spectral Filtering

Traditional signal processing method.

Fast but limited.

Neural Speech Enhancement

Deep learning models perform:

- noise suppression

- dereverberation
- signal reconstruction.

Examples:

Demucs
DeepFilterNet
RNNoise.

Evaluation Metrics

Enhancement quality measured using:

PESQ (Perceptual Evaluation of Speech Quality)

STOI (Speech Intelligibility Index)

SI-SDR (Signal-to-Distortion Ratio).

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8. Speaker Separation

When multiple people speak simultaneously, the system must isolate individual voices.

This problem is known as **source separation**.

Deep Learning Solutions

Conv-TasNet

Time-domain speech separation.

High performance but computationally intensive.

Demucs

Hybrid waveform-spectral model.

Strong separation performance.

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9. Streaming Speech Recognition

Speech recognition converts audio into text.

Key challenge:

ASR must operate **in real time**.

9.1 Traditional Models

HMM-DNN systems

Older architecture.

Limited performance compared to modern models.

9.2 End-to-End Models

Modern ASR systems use neural architectures.

Examples:

RNN-T
Conformer
Transformer.

9.3 Streaming Constraints

Standard transformers require full sequence context.

This prevents real-time inference.

Emformer

Proposed by Facebook AI.

Introduces **memory-efficient streaming attention**.

Key innovation:

Segment-based attention with memory.

Benefits:

- streaming capability
- bounded memory
- low latency.

Reference:

Shi et al., *Emformer: Efficient Memory Transformer for Streaming Speech Recognition*.

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10. Speaker Diarization

Goal:

Determine **who spoke when**.

Approach:

1. Speaker embedding extraction
2. Clustering of embeddings.

Tools:

pyannote.audio
NVIDIA NeMo.

Evaluation metric:

Diarization Error Rate (DER).

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11. Multi-Resolution Signal Processing

This is the **core conceptual contribution of your project**.

Why multi-resolution matters

Speech signals contain information across multiple scales.

Examples:

Short timescale

Phonemes.

Long timescale

Prosody and sentence structure.

Frequency domain

Different frequency bands carry different speech features.

Limitations of Single-Resolution Features

Standard ASR uses:

Mel spectrograms.

These capture fixed-scale representations.

Limitations:

- limited temporal resolution
- sensitive to noise.

Multi-Resolution Analysis

Uses multiple time-frequency representations.

Examples:

Wavelet transform
Multi-band spectral analysis
Multi-resolution CNNs.

Benefits

Improved robustness

Noise often affects specific frequency bands.

Multi-resolution processing allows selective filtering.

Interpretability

Wavelet coefficients provide interpretable representations of signal structures.

Efficient representation

Hierarchical feature extraction captures signal characteristics more efficiently.

12. Energy-Efficient System Architecture

Continuous listening requires careful power management.

Event-driven inference

Most of the time, no speech is present.

Tiny models detect speech activity.

Heavy models activate only when needed.

Cascaded Models

Architecture:

Tiny model

↓

Medium model

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Large model

Each stage increases computational complexity.

Model compression techniques

Quantization

Reduce numerical precision.

Example:

INT8 inference.

Knowledge Distillation

Train small models to mimic large models.

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13. Datasets for Validation

Speech recognition datasets:

LibriSpeech
Common Voice
VoxPopuli.

Noise datasets:

MUSAN
DNS challenge.

Separation datasets:

LibriMix
WSJ0-mix.

14. Evaluation Metrics

Define evaluation metrics clearly.

Speech recognition:

Word Error Rate (WER).

Speech enhancement:

PESQ
STOI.

Localization:

Angular error.

Speaker diarization:

DER.

System efficiency:

Latency
Energy consumption.

15. Experimental Validation Strategy

Experiments should evaluate each subsystem.

Stage 1

Compare localization algorithms.

Stage 2

Evaluate beamforming techniques.

Stage 3

Compare speech enhancement models.

Stage 4

Evaluate streaming ASR architectures.

Stage 5

Test full integrated pipeline.

Sound Source Localization: Technical Deep Dive and Design Rationale

To develop a robust and deployable sound source localization (SSL) module for edge-based audio intelligence systems, we conducted an extensive study of both classical and deep learning-based localization frameworks, supported by controlled simulation experiments conducted in a custom-built acoustic simulation lab. This section summarizes the key technical insights obtained from our investigation and presents the reasoning behind the final system design.

1. Foundations of Sound Source Localization

Sound source localization (SSL) refers to the estimation of the spatial origin of sound relative to a microphone array, typically simplified as estimating the direction of arrival (DoA) in azimuth and elevation. Localization relies fundamentally on interchannel differences between microphone signals, particularly time delays, phase shifts, and amplitude variations introduced by propagation geometry and room acoustics

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In real environments, each microphone receives a convolution of the source signal with a room impulse response (RIR), incorporating both direct paths and reflections. Consequently, SSL becomes a highly complex inverse problem because the source signal, propagation delay, and acoustic channel response are all unknown and must be inferred jointly

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2. Classical SSL Approaches and Limitations

We first examined traditional signal processing-based SSL methods, which remain foundational.

2.1 GCC-PHAT and TDoA Methods

GCC-PHAT estimates the time difference of arrival (TDoA) between microphone pairs using phase-weighted cross-correlation. The PHAT weighting removes amplitude information and emphasizes phase consistency, improving robustness in reverberant conditions

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However, these approaches:

• degrade significantly in reverberant environments
• struggle with multiple simultaneous sources
• require explicit geometry modeling.

2.2 SRP-PHAT (Beamspace Search)

SRP-PHAT extends GCC by scanning possible spatial directions and computing an acoustic power map. Localization is obtained from the peak response

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Advantages:

• strong robustness
• multi-microphone integration

Limitations:

• computationally expensive (grid search across directions)
• difficult for real-time edge deployment.

2.3 Subspace Methods (MUSIC)

Subspace methods analyze the spatial covariance matrix of microphone signals and use eigenvalue decomposition to separate signal and noise subspaces

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Key insight:

Localization is obtained when steering vectors align with the signal subspace and remain orthogonal to the noise subspace.

While highly accurate under ideal conditions, MUSIC suffers from:

• sensitivity to reverberation
• computational cost (eigen decomposition)
• limited scalability to multiple sources.

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3. Deep Learning-Based SSL: Core Advances

Deep learning methods address limitations of classical techniques by learning the complex mapping between multichannel acoustic patterns and spatial coordinates.

3.1 Physics-Informed Neural SSL

Modern approaches retain physical priors while leveraging learning-based inference:

Pipeline:

Multichannel signal → STFT → covariance matrix → eigenvectors → neural mapping

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Instead of explicitly computing MUSIC spectra, networks learn how eigenvector structures correspond to spatial directions.

This provides:

• improved robustness in noisy environments
• better multi-source handling
• reduced computational complexity per frame.

3.2 Multi-Source Localization and WDO Principle

A critical insight from literature is **W-disjoint orthogonality (WDO)**: speech signals are sparse in the time-frequency domain, meaning individual TF bins are typically dominated by one speaker

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This allows:

• independent DoA estimation per frequency bin
• effective separation of overlapping sources.

3.3 Joint Localization and Activity Detection

Modern SSL systems integrate:

• voice activity detection (VAD)
• source counting
• localization.

Designs such as ACCDOA encode both activity and direction in a single representation, improving efficiency

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4. Simulation Acoustic Lab (Our Implementation Contribution)

A major challenge in SSL research is the lack of labeled real-world data due to difficulty obtaining ground-truth spatial annotations. Consequently, modern SSL systems rely heavily on simulated acoustic datasets

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To address this, we developed a **simulation acoustic lab** where:

• room impulse responses were generated using the image source method
• multiple room geometries and reverberation profiles were simulated
• diverse source positions and microphone configurations were evaluated.

Synthetic multichannel signals were created by convolving dry speech signals with simulated RIRs and augmenting them with realistic environmental noise

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This allowed controlled evaluation across:

• varying SNR levels
• multiple speakers
• dynamic spatial configurations.

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5. Efficient CNN-Based SSL for Edge Deployment

The most practical design direction emerged from CNN-based architectures optimized for edge computing.

Key innovation: Multi-Stream Block

Parallel convolution filters (kernel sizes 3, 5, 7) capture multi-scale delay structures simultaneously

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Insights:

• small kernels capture fine phase shifts
• large kernels capture broader delays
• aggregation gates adaptively fuse scales.

This achieves multiresolution feature extraction while minimizing depth and computation.

Efficiency Enhancements

The literature identifies several optimizations:

• depthwise separable convolutions reduce computation by ~25%

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• stride-based downsampling preserves phase better than pooling
• hybrid coarse-classification + fine-regression reduces dense-layer cost.

These strategies enabled real-time performance (~7.8 ms per 40 ms frame on Raspberry Pi)

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6. Feedback Loop Between Localization, Separation, and Diarization

An important conceptual insight from the study is the strong coupling between localization, diarization, and source separation.

Localization supports separation by identifying spatially distinct sources, while diarization supports localization by maintaining speaker identity across time, even during motion

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Tracking frameworks combine temporal consistency with spatial estimation, forming a closed-loop perception system.

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7. Final Design Choice and Justification

Based on theoretical study and simulation experiments, the final design direction prioritizes:

• CNN-based multi-stream SSL
• physics-informed feature learning
• synthetic-data-driven training
• lightweight architecture for edge deployment.

Compared to classical methods, this approach offers:

✔ stronger robustness to noise and reverberation
✔ improved multi-source handling
✔ significantly lower computational complexity
✔ suitability for real-time embedded deployment.